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英集芯科技 INJOINIC TECHNOLOGY

IP6525T

Maximum output 18W, integrated fast charging output protocol

(DCP/QC2.0/QC3.0/FCP/AFC) buck SOC

1 Features

ÿ Synchronous switching buck converterÿ
Built-in power MOSÿ Input voltage range: 9.6V to 32Vÿ Output voltage range: 3V to 12V, automatically adjusted according to the fast charging protocolÿ Output power: Maximum 18W (5V@3.4A, 9V@2A, 12V@1.5A, etc.)ÿ Output has CV/CC characteristics (output current is less than the set value, output CV mode; output current is greater than the set value, output CC mode)ÿ Conversion efficiency up to 98%ÿ Soft start functionÿ Output voltage line compensation function, line compensation voltage 50mV@1A

ÿ Output fast charge ÿ
Support BC1.2, Apple, Samsung protocols ÿ Support Qualcomm QC2.0 and QC3.0 ÿ Support Huawei fast charge protocol FCP ÿ Support Samsung fast charge protocol AFC

ÿ Multiple protections, high reliability
ÿ Output overcurrent, input overvoltage, input undervoltage, output short circuit protectionÿ Whole machine overtemperature protectionÿ DP/DM overvoltage protectionÿ ESD 4KV, input DC withstand voltage 40V

2 Applications

ÿCar chargerÿQuick charging adapterÿSmart power strip

3 Introduction

IP6525T is a buck converter with integrated synchronous switch, supporting 4 output fast charging protocols, providing a complete solution for car chargers, fast charging adapters, and smart power strips.

IP6525T has built-in power MOS, input voltage range is 9.6V to 32V, output voltage range is 3V to 12V, maximum output power is 18W, and output voltage and current are automatically adjusted according to the fast charging protocol. Typical output voltage and current are 5V@ 3.4A, 9V@2.0A, 12V@1.5A. The buck conversion efficiency of IP6525T is as high as 98%.

The output of IP6525T has CV/CP/CC characteristics. When the current is less than the set value, the output is in CV mode and the output voltage is constant; when the output current is greater than the set value, the output is in CP mode, the load increases, and the output voltage decreases; when the voltage drops to the set voltage value, the output In CC mode, as the load increases, the output voltage decreases rapidly until the output undervoltage protection is triggered.

The output voltage of IP6525T has a line compensation function. When the output current increases, the output voltage will increase accordingly to compensate for the voltage drop caused by the connection line impedance.

IP6525T has a soft start function, which can prevent the inrush current at startup from affecting the stability of the input power supply.

IP6525T integrates fast charging protocol, which can automatically identify the fast charging protocol supported by the output terminal access device through DP/DM, and then automatically adjust the output voltage and current. IP6525T supports fast charging protocols such as DCP (Apple, Samsung and BC1.2), Qualcomm QC2.0/QC3.0, Huawei fast charging protocol FCP, Samsung fast charging protocol AFCÿ

IP6525T has multiple protection functions, including input overvoltage, undervoltage protection, output overcurrent, overvoltage, undervoltage, short circuit protection, etc.

IP6525T adopts ESOP8 package.

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4 Pin Definition

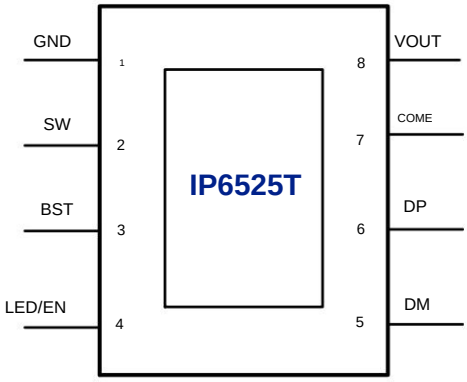


Figure 2 IP6525T pin diagram

Serial number	name	describe
1	GND	Power ground
2	SW	DCDC switch node, connect inductor
3	BST	The bootstrap circuit pins are placed close to the chip BST pin and SW pin to place the bootstrap capacitor for the upper tube Gate drive provides voltage
4	HLED/EN	fast charge LED display pin/external enable (EN function must be customized)
5	DM	USB fast charge identification signal DM
6	DP	USB fast charge identification signal DP
7	VIN	input voltage pin, close to the IC, a filter ceramic capacitor needs to be placed, 100nF is recommended
8	VOUT	Output voltage feedback pin
9(EPAD)	GND	Power ground and heat dissipation ground, need to maintain good contact with GND

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5. IP6525T model description

Model **NameUSB**	Output Power	**PIN4** function setting
IP6525T	USBA QC(1) 5V/3A 9V/2A 12V/1.5A	HLED function. If this function is not needed, this PIN is left floating.
IP6525T_EN	USBA QC(1) 5V/3A 9V/2A 12V/1.5A	EN function. If this function is not needed, this PIN is left floating.

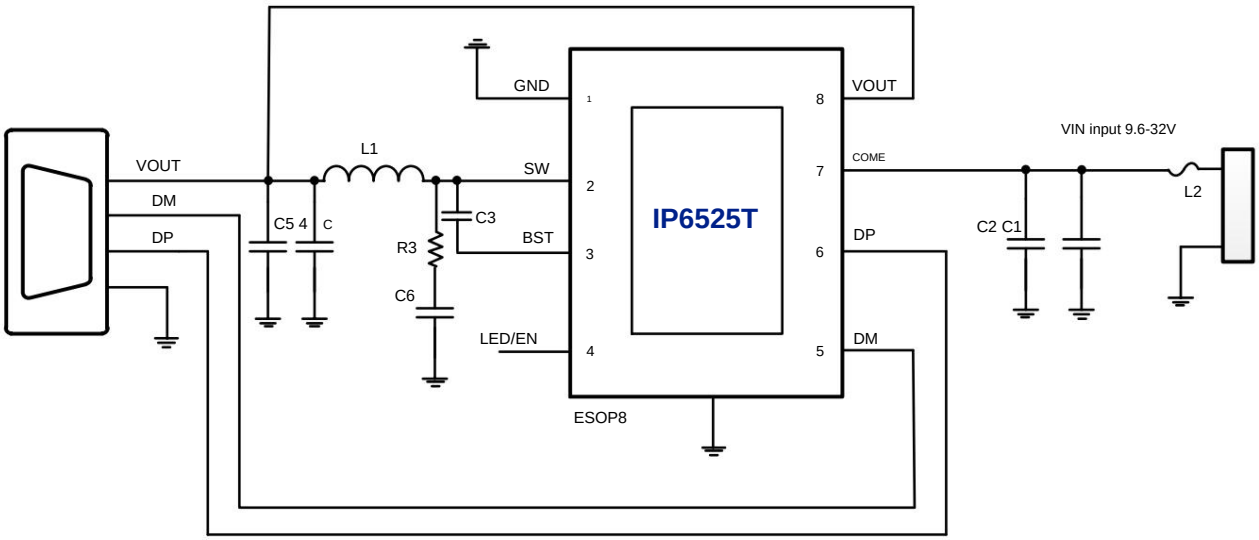


Figure 1 IP6525T simplified application schematic

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6 Limit parameters

parameter	symbol	value	unit
Input voltage range	COME	-0.3 ~ 40	**In**
SW voltage range	VSW	-0.3 ~ 40	**In**
DM/DP voltage range	In DM/DP	-0.3 ~ 6	**In**
HLED/EN Voltage Range	In IN	-0.3 ~ 6	**In**
VOUT Voltage Range	In VSP/VSN	-0.3 ~ 25	**In**
Junction temperature range	TJ	-40 ~ 150	ÿ
Storage temperature range	Test	-60 ~ 150	ÿ
Thermal resistance (junction to ambient)	ÿJA	40	**ÿ/W**
Human Body Model (HBM)	ESD	4	**KV**

* Stresses above those listed under the Absolute Maximum Ratings section may cause permanent damage to the device.

Excessive exposure time may affect the reliability and service life of the device

7 Recommended working conditions

parameter	symbol	Minimum	Typical Value	Maximum	unit
Input voltage	COME	9.6	12/24	32	In

*Device operational characteristics are not guaranteed outside these operating conditions.

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8 Electrical characteristics

Unless otherwise specified, TA=25ÿ, L=22uH, COUT=220uF E-cap (About 100mÿ ESR), VIN=12V, VOUT=5V

parameter	symbol	Test conditions	Min	Typ	Max	Unit
Input System						
Input voltage	COME		9.6	12	32	In
Input undervoltage threshold VIN-UV		Rising voltage		9.6		In
		Drop voltage		9.1		In
Input overvoltage threshold VIN-OV		Rising voltage		32		In
		Recovery hysteresis voltage		0.2		In
Input quiescent current IQ		VIN=12VÿVOUT=5V@0A		1.5		m.a.
Power Switching System						
Upper tube on-resistance RDS(ON)				30		mÿ
Lower tube on-resistance RDS(ON)				20		mÿ
Switching frequency	FSW			110		KHz
Output system						
Output voltage	VOUT		3	5	12	In
Output voltage ripple ÿVOUT		VIN=12VÿVOUT=5V@3A		135		mV
		VIN=12VÿVOUT=9V@2A		75		mV
		VIN=24VÿVOUT=12V@1.5A		180		mV
		Note: Typical values tested based on the demo board reference design				
Soft start time TSS		VIN=12VÿVOUT=5V		1.4		ms
Output line compensation voltage VCOMP		VIN=12V, VOUT=5V, IOUT=3A		150		mV
Output CC mode Maximum current	IOUT	VIN=12VÿVOUT<=4V		3.4		A
		VIN=12Vÿ4V<VOUT<=5V		3.4		A
		VIN=12Vÿ7V<VOUT<=9V		2		A
		VIN=24Vÿ9V<VOUT<=12V		1.5		A
Output hiccup restart Voltage	VOUT	After the output enters CC mode, the output hiccup resets Start voltage		2.6		In
Thermal shutdown temperature TOTP rising temperature				150		ÿ
Thermal shutdown delay Hysteresis	ÿTOTP			40		ÿ

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9 Functional Description

9.1 Synchronous Switching Buck Converter

IP6525T integrates a synchronous switching buck converter. The input voltage range is 9.6V~32V, and the output voltage range is 3V~12V. Output 5V/3.4A, 9V/2A, 12V/1.5A. IP6525T has a built-in power switch tube, and the switching frequency during operation is 110KHz. When VIN=12V, VOUT=5V@3A, the conversion efficiency is 95.5%. The efficiency curve is shown in Figure 3. Figure 4 shows the IP6525T output Output voltage and load current relationship curve.

IP6525T can automatically adjust the output voltage based on the identified fast charging protocol.

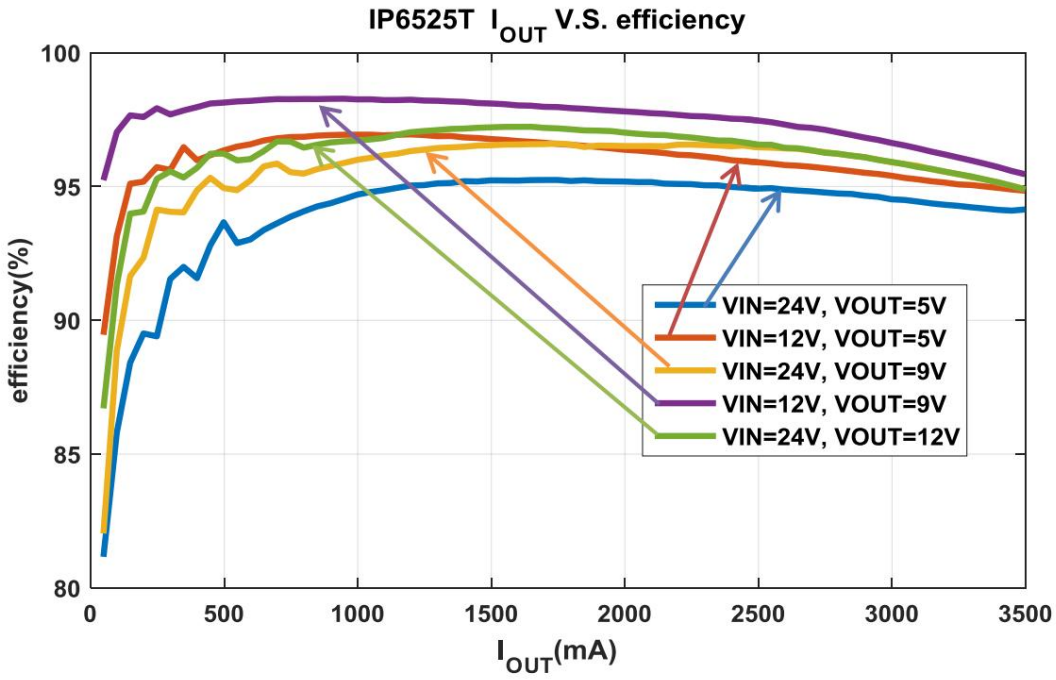


Figure 3 IP6525T efficiency curve

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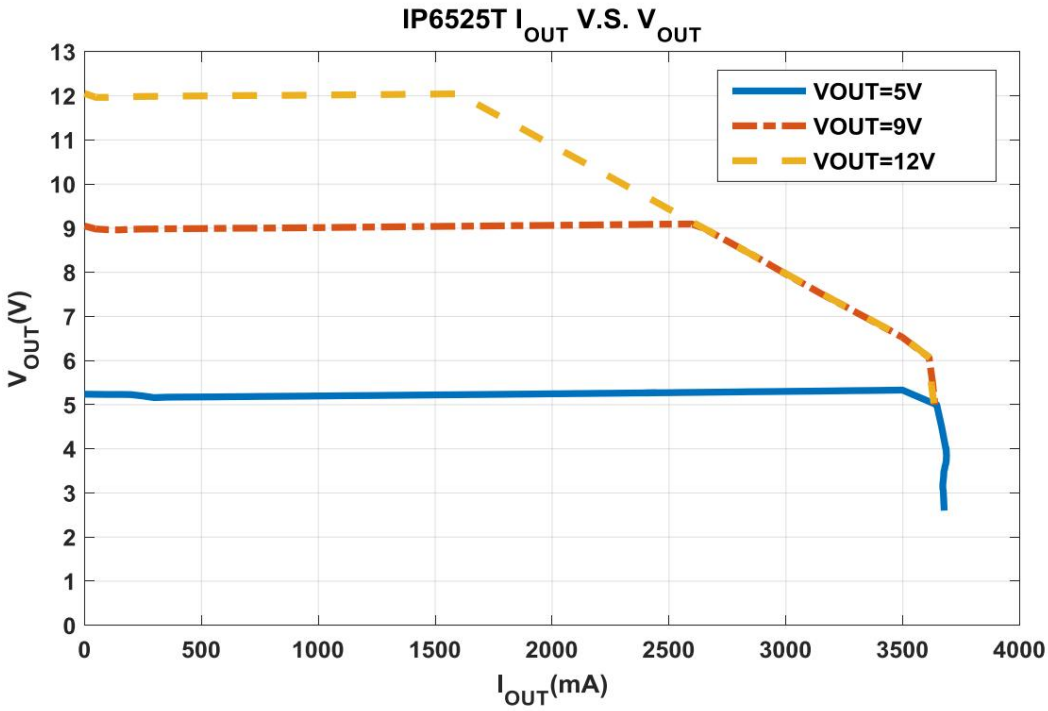


Figure 4 IP6525T output voltage and current relationship curve

9.2 Output voltage line compensation function

IP6525T has a wired output voltage compensation function: when the output current is 3A, the output voltage will increase by 150mV.

9.3 External Enable Function

PIN-4 can be customized as an external enable function and used as an EN PIN. When it is pulled low externally, the DCDC output can be turned off.

When used as EN enable, this PIN cannot be directly connected to the high voltage VIN, otherwise it will cause overvoltage breakdown of this PIN-4.

9.4 Fast charging **LED** display function

PIN-4 can be customized as a fast charge LED display function. It can be used as an HLED and can be used by connecting an external light-emitting diode. It will light up when the fast charge request voltage is not 5.0V, that is, it will light up when the request voltage is higher or lower than 5.0V.

9.5 Output **CC/CP/CV** Characteristics

The output of IP6525T has CV/CP/CC characteristics: when the output current is less than the set value, the output is in CV mode and the output voltage is constant; when the output current is greater than the set current limit value, it enters CP mode, and as the load increases, the output voltage decreases; when the voltage drops to 6V, it enters CC mode, the load continues to increase, and the output voltage decreases rapidly until the output voltage undervoltage protection is triggered;

9.6 Protection Function

IP6525T will detect the VIN voltage. If the VIN voltage is lower than 9.6V, IP6525T will enter standby mode and shut down the output.

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IP6525T has input overvoltage protection function: when VIN rises to 32V, IP6525T detects input overvoltage and shuts down the output; when VIN drops When the voltage is about 0.2V lower, IP6525T considers the input to be normal and turns on the output.

IP6525T has output undervoltage protection function: when VOUT output drops to 2.5V, IP6525T detects output undervoltage and turns off the output. Hiccup restarts after 2s.

IP6525T has short-circuit protection function. If the VOUT voltage is lower than 2.6V after 8ms of startup, IP6525T will think that the output is short-circuited and shut down. Output, hiccup and restart after 2s.

IP6525T has over-temperature protection function: when IP6525T detects that the chip temperature reaches 150ÿ, it will shut down the output and restart after 2s. When the temperature drops to 110°C, IP6525T considers that the temperature has returned to normal and turns on the output again.

9.7 Output fast charging protocol

IP6525T supports multiple output fast charging protocols:

ÿ Support DCP protocol (Apple, Samsung and BC1.2)

ÿ Support Qualcomm QC2.0 and QC3.0

ÿ Support Huawei fast charging protocol FCP

ÿ Support Samsung fast charging protocol AFC

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10 Typical application schematic diagram

The IP6525T only requires a few inductors, capacitors, and resistors to implement a fully functional car charging solution.

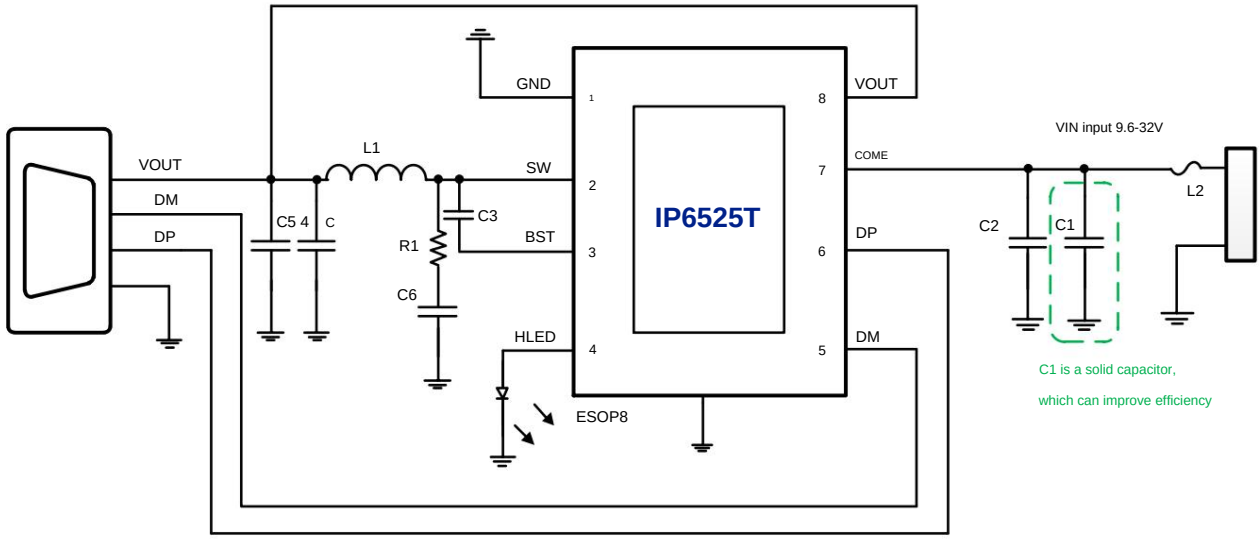


Figure 5 IP6525T output application schematic

illustrate:

1. The EPAD of IP6525T must be in good contact with the GND of the PCB board.

2. C2 should be placed close to PIN7 of the device;

3. If the HLED indicator light is not needed, PIN4 cannot be connected to anywhere and can be left floating.

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11 BOM

Serial number component name	Models & Specifications	Unit usage location			Remark
1 IC	IP6525T	PCS 1		U1	
2 electrolytic capacitors 100uF/35V		PCS 1		C1	The withstand voltage is greater than 35V. Using solid capacitors can improve efficiency.
3 electrolytic capacitors 220uF/25V		PCS 1		C4 withstand voltage value greater than 25V	
4 TC-220M-4.5A-CS137125	22uH+/-20%, current 4.5A DCR<12mohm	PCS 1		L1 3L Electronic	
5 SMD capacitor 0603 100nF 10%		PCS 1		C2	The withstand voltage is greater than 35V, which is required for layout Place it close to the IC PIN
6 SMD capacitor 0603 100nF 10%		PCS 2		C3 C5	Withstand voltage greater than 16V
7 Chip resistor 0603 2ohm 5%		PCS 1		R1	
8 SMD capacitor 0603 1nF 10%		PCS 1		C6	
9 Fuse F1		PCS 1		F1 current value 4A	
10 SMD LED	0603	PCS 1		D1	

Recommended inductor model: TC-220M-4.5A-CS137125

3L product No.	Inductance (uH)	Tolerance	DC Resistance (mÿ)		Heat Rating Current DC Amp.	Saturation Current DC Amps.	Measuring Condition
			Type.	Max.	Idc(A)Max	Isat(A)Max	
TC-220M-4.5A-CS137125	22.0	±20%	12	14	4.5	8	

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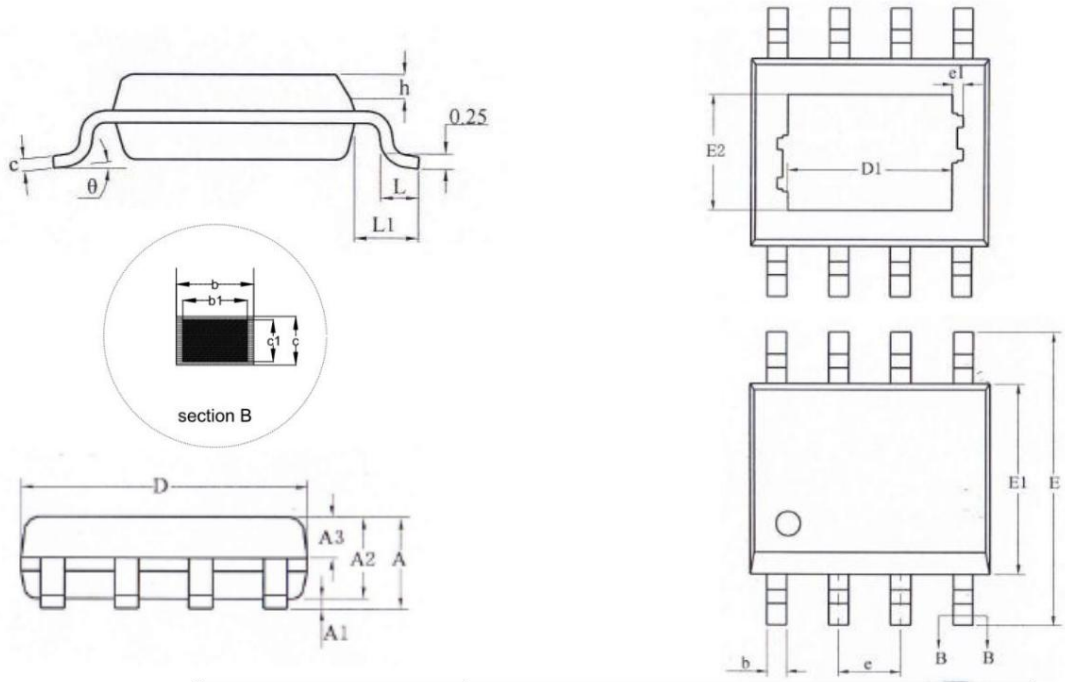
12. IP series car charger IC model selection table

IC Model	Discharge current	pair road	Supported protocols										Encapsulation	
			DCP	QC2.0	QC3.0	FCP	SCP	AFC		MTK ON	SFCP	PD2.0 PD3.0 (PPS)	Specification	Allow
IP6523S_N	3.4A	-	ÿ	-	-	-			-	-	-	-	ESOP8	
IP6536	2.4A	ÿ ÿ		-	-	-			-	-	-	-	ESOP8	
IP6525T	18W	-	ÿ ÿ		ÿ	ÿ	- ÿ	-			-	-	ESOP8	
IP6525S	18W	-	ÿ ÿ		ÿ	ÿ	ÿ ÿ	ÿ			ÿ	-	ESOP8	
IP6510	18W	-	ÿ ÿ		ÿ	ÿ	- ÿ	-			-	ÿ	ESOP8	
IP6520	18W	-	ÿ ÿ		ÿ	ÿ	ÿ ÿ	ÿ			-	ÿ	ESOP8	
IP6520_PPS	18W	-	ÿ ÿ		ÿ	ÿ	ÿ ÿ	ÿ			-	ÿ	ÿ ESOP8	
IP6537_C	18W	-	ÿ ÿ		ÿ	ÿ	ÿ ÿ	ÿ			ÿ	ÿ	ÿ QFN24	
IP6537_C_30W20V	30W	-	ÿ ÿ		ÿ	ÿ	ÿ ÿ	ÿ			ÿ	ÿ	ÿ QFN24	
IP6515	4.8A	ÿ ÿ		-	-	-			-	-	-	-	QFN32	
IP6538_CC	27W	ÿ ÿ ÿ				ÿ	ÿ - ÿ	ÿ			-	ÿ	ÿ QFN32	
IP6538_AC	27W	ÿ ÿ ÿ				ÿ	ÿ ÿ	ÿ ÿ			-	ÿ	ÿ QFN32	
IP6538_AA	24W	ÿ ÿ ÿ				ÿ	ÿ ÿ	ÿ ÿ			-	-	QFN32	
IP6527S_A	24W	-	ÿ ÿ		ÿ	ÿ	ÿ ÿ	ÿ ÿ			-	-	QFN32	
IP6527S_C	27W	-	ÿ ÿ		ÿ	ÿ	- ÿ	ÿ			-	ÿ	ÿ QFN32	
IP6527S_C_18WPD	18W	-	ÿ ÿ		ÿ	ÿ	- ÿ	ÿ			-	ÿ	ÿ QFN32	

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13 Packaging Information



SYMBOL	MILLIMETER		
	MIN	NAME	MAX
A	-	-	1.65
A1	0.05	-	0.15
A2	1.30	1.40	1.50
A3	0.60	0.65	0.70
b	0.39	-	0.47
b1	0.38	0.41	0.44
c	0.20	-	0.24
c1	0.19	0.20	0.21
D	4.80	4.90	5.00
AND	5.80	6.00	6.20
E1	3.80	3.90	4.00
and	1.27BSC		
h	0.25	-	0.50
L	0.50	0.60	0.80
L1	1.05REF		
i	0	-	8th
D1	-	3.10REF	-
E2	-	2.21REF	-

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